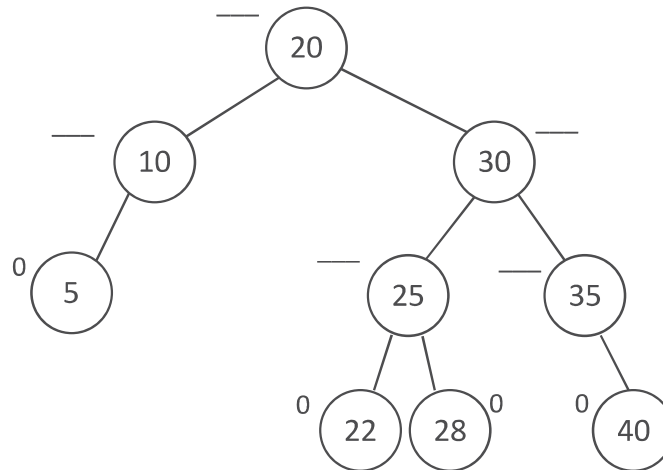


1. (12 pts.) **AVL trees:** Consider the AVL tree shown below.



- 1.1. Choose the correct balance for the node with key **10**? [ -2 / -1 / 0 / +1 / +2].
- 1.2. Choose the correct balance for the node with key **20**? [ -2 / -1 / 0 / +1 / +2].
- 1.3. Choose the correct balance for the node with key **25**? [ -2 / -1 / 0 / +1 / +2].
- 1.4. Choose the correct balance for the node with key **30**? [ -2 / -1 / 0 / +1 / +2].
- 1.5. Choose the correct balance for the node with key **35**? [ -2 / -1 / 0 / +1 / +2].
- 1.6. Which node, if any, when removed would necessitate a rotation(s)? [ List the node's key or type "none" if none exists].
- 1.7. Show the process for **insert(29)** including where it is added, updates of any balance values, and the result of any necessary rotations. You can draw the tree at a few points in time during the insert process and need only draw the relevant parts of the tree (though you must show the relevant parts). Upload a PDF or image of your work.